

Design Pattern 3: Builder Pattern

Exercise 3: Implementing the Builder Pattern

Scenario:

You are developing a system to create complex objects such as a Computer with multiple optional parts. Use the Builder Pattern to manage the construction process.

Steps:

Create a New Java Project:

Create a new Java project named BuilderPatternExample.

Define a Product Class:

Create a class Computer with attributes like CPU, RAM, Storage, etc.

Implement the Builder Class:

Create a static nested Builder class inside Computer with methods to set each attribute.

Provide a build() method in the Builder class that returns an instance of Computer.

Implement the Builder Pattern:

Ensure that the Computer class has a private constructor that takes the Builder as a parameter.

Test the Builder Implementation:

Create a test class to demonstrate the creation of different configurations of Computer using the Builder pattern.